INCH-POUND

MIL-P-71158 (AR)
19 October 1994

MILITARY SPECIFICATION

PROPELLANT, LIQUID, XM46

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the manufacture and packaging of XM46 liquid propellant.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army, ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1376

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MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1171	- Acceptance and Description Sheets (for Propellants and Explosives)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY ARMAMENTS RESEARCH DEVELOPMENT AND ENGINEERING CENTER

12972309	- Packing and Marking for XM46 Liquid Propellant Shipping & Storage Container (30 Gallon Drum)
12972308	- Marking for XM46 Liquid Propellant Inner Pack(5 Gallon Drum)
12972305	- Drum, Shipping, Inner for XM46 Liquid Propellant (5 Gallon Drum)
12972315	- Packing and Marking for XM46 Liquid Propellant Shipping and Storage Container (55 Gallon Drum)
12972314	- Marking for XM46 Liquid Propellant Inner Pack (30 Gallon Drum)
12972311	- Drum, Shipping, Inner for XM46 Liquid Propellant (30 Gallon Drum)

(Copies of drawings, publications and other Government documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 203	- Test Method for Water Using Karl Fischer Reagent

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ASTM E 300	- Recommended Practice for Sampling Industrial Chemicals
ASTM D 2905	- Statement on Number of Specimens Required to Determine the Average Quality of a Textile Material

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN CHEMICAL SOCIETY SPECIFICATIONS

Reagent Chemicals

(Applications for copies should be addressed to the American Chemical Society, 1155 16th Street NW, Washington, DC 20036)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Chemical and physical properties. XM46 liquid propellant shall be a mixture of 60.8 wt % Hydroxyl ammonium nitrate (HAN), 19.2 wt % Triethanol ammonium nitrate (TEAN) and 20.0 wt % water. The propellant shall conform to the chemical and physical requirements specified in Table I when tested in accordance with the applicable paragraph.

TABLE I
Chemical and Physical Requirements

	Requirement		Applicable
	Min	Max	Test Method
HAN, wt %	60.6	61.0	4.5.1.1
TEAN, wt %	19.0	19.4	4.5.1.1
Water, wt %	19.8	20.2	4.5.1.2
Nitric acid, wt %	0.001	0.05	4.5.1.3
Ammonium Nitrate, wt %	---	0.50	4.5.1.4
Iron, ppm	---	2.0	4.5.1.5
Mercury, ppb	---	200.	4.5.1.5
Total Metals, ppm	---	5.0	4.5.1.5
Density, gm/cc	1.410	1.450	4.5.1.6
Sulfate ions, ppm	---	10.0	6.8

3.1.1 Metals. The following metals shall be included in the value for total metals: Chromium, Cobalt, Copper, Iron, Manganese, Nickel, Vanadium, Zinc and Mercury. Their sum shall not exceed 5.0 ppm. Furthermore, the concentration of Mercury

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shall not exceed 200. ppb and the concentration of Iron shall not exceed 2.0 ppm (See 6.6).

3.1.2 Closure. The sum of concentration of HAN, TEAN, water and AN shall add up to 100 - 0.6 + 1.1 wt %.

3.1.3 Total nitrate. The total nitrate concentration of the propellant shall be determined in accordance with 4.5.1.7 for information purposes.

3.2 Form. The XM46 propellant shall be in the form of a clear, colorless liquid.

3.3 First article inspection. When specified in the contract or purchase order (see 6.2) a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.4 Propellant description sheets. The contractor shall submit a propellant description sheet and a propellant acceptance sheet in accordance with MIL-STD-1171. The sheets, as a minimum, will provide the necessary information in order to demonstrate that the propellant meets the requirements of this specification. These sheets shall be submitted to the Government in accordance with 6.9.

3.5 Workmanship. The manufacturer shall implement procedures and controls to assure that the process and the product produced are not compromised by foreign materials and contaminants or any other conditions which may degrade the composition. The propellant shall have no detectable odor during normal handling operations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products and supplies submitted to the Government for acceptance

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comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. References shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall be 500 gm of XM46 propellant for chemical testing. A sample of propellant for ballistic testing will be submitted in accordance with the provisions of the contract. The sample shall be obtained from the first production lot which has been produced by the Contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production. In addition, a new sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in the manufacturing process, material used, specification or source of supply as to significantly affect the product as determined by the contracting officer. The contractor is in no way authorized by the Government to initiate production of a second lot of propellant until the first article test has been completed and the results are found to be acceptable.

4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in Table II (MIL-A-48078).

4.3.3 Rejection. If any test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of a test specimen to comply with any of the requirements.

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER / NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	HAN, wt %		3.1	4.5.1.1
	TEAN, wt %		3.1	4.5.1.1
	Water, wt %		3.1	4.5.1.2
	Nitric acid, wt %		3.1	4.5.1.3
	Ammonium nitrate, wt %		3.1	4.5.1.4
	Total metals, ppm		Note 1	4.5.1.5
	Mercury, ppb		3.1	4.5.1.5
	Total impurities, wt %		3.1	4.5.1.6
	Density, gm/cc		3.1	4.5.1.7
	Form		3.2	4.5.1.9
	Ballistics		Note 2	
	Thermal stability			6.7
	Sulfate, ppm		3.1	6.8

NOTES:

Note 1 - In addition the metals required in 3.3, the metals called out in 6.7 shall be determined.

Note 2 - Requirements for Ballistics will be specified in the contract..

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Unless otherwise approved by the Contracting officer, a lot of XM46 liquid propellant shall be no less than 5,000 Kg. In addition the following criteria shall apply: A lot shall consist of one or more batches of propellant produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of XM46 propellant that has been blended and was subjected to the same unit chemical or physical process intended to make the final product homogeneous. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168. Also, MIL-A-48078 shall apply. Furthermore, a lot of propellant shall contain:

a. Hydroxyl ammonium nitrate from not more than one manufacturer using the same manufacturing process.

b. Triethanol ammonium nitrate from not more than one manufacturer using the same process.

4.4.1.1 Blending of batches. When batches of propellant are blended to form a lot, all batches that are to make up the lot must simultaneously be placed in one vessel and mixed until homogeneous or the contractor must receive prior approval on a blending procedure if the condition above is not met (See 6.5).

4.4.2 Examination. See MIL-A-48078.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawings and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table III below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE III
Attributes Sample Inspection

Lot Size	Major Sample Size	Minor Sample Size
2 to 8	*	See Note
9 to 15	*	5
16 to 25	*	5
26 to 50	32	5
51 to 90	32	12
91 to 150	32	12
151 to 280	32	14
281 to 500	32	17
501 to 1200	74	20
1201 to 3200	74	24
3201 to 10,000	87	30
10,001 to 35,000	109	36
35,001 to 150,000	124	40
150,000 to 500,000	156	40
500,001 and over	187	40

* - 100 % inspection

NOTE: Use 100 % inspection when lot size is less than 5 units. Use sample size when lot size is greater than 5 units.

b. Alternate quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.11). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Drum, Shipping, Inner for XM46 Liquid Propellant (5 Gallon/30 Gallon) (Prior to Filling)	SHEET 1 OF 1	12972305/12972311
			NEXT HIGHER ASSEMBLY 12972308/12972314

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Foreign material in drum.	100 %	5.1	Visual
102	Gasket missing or improperly attached.	100 %	5.1	Visual
Minor	None defined.			

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Marking for XM46 Liquid Propellant Inner Pack (5 Gallon Drum/ 30 Gallon Drum) (After Filling)	1 OF 1	12972308/12972314
			NEXT HIGHER ASSEMBLY 12972309/12972315

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Container damaged to the extent propellant can escape.	100 %	5.1	Visual
102	Weight of loaded drum.	100 %	5.1	Visual
103	Presence of Leak Detector & Paper	100 %	5.1	Visual
Minor				
201	Marking misleading or unidentifiable.	Table III	5.3	Visual

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4.4.3 Testing. PRECAUTION: This specification covers sampling and testing of toxic and hazardous material. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing this propellant.

4.4.3.1 Sampling for chemical/physical testing. Approximately 250 grams of the propellant shall be selected for each sample from each lot using ASTM Method E300 for liquids. Samples shall be drawn from the top, middle and bottom of the holding tank or reaction vessel prior to loading into drums or from the first, middle and last drum during the drumming operation. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected (note). Each required test shall be performed in triplicate. No composite samples shall be used. The classification of defects shall be major unless otherwise specified.

Note - If one of the three samples fails to meet any of the requirements, three new samples may be drawn from the tank or drums previously sampled and retest may be performed on all three samples. If all of the three samples meet the requirement of the test that initially failed, the lot represented by the sample will be accepted. The result of the retests and the initial results of the assay shall be documented on the propellant description sheet.

4.4.4 Inspection equipment. For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection equipment should be employed. The contractor shall have available and utilize correctly, this equipment and is charged with the responsibility of ensuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.3).

4.5 Methods of inspection.

4.5.1 Chemical and physical testing. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall meet the American Chemical Society (ACS) standard for reagent chemicals specified in the current edition of Reagent Chemicals. See 6.4 for the use of equivalent methods.

4.5.1.1 HAN and TEAN concentration.

4.5.1.1.1 Equipment.

a. Automatic titrator, 716 DMS Titrino or equivalent.

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b. Magnetic stirrer.

c. pH meter.

4.5.1.1.2 Reagents.

a. Acetone.

b Standardized Sodium hydroxide (NaOH) solution, 0.2 N (measured to four decimal places).

4.5.1.1.3 Procedure.

a. Dilute 0.55 to 0.60 gm of propellant with 80 ± 1 ml of distilled water.

b. Add 5 ± 0.1 ml of acetone to the solution and allow ten minutes under gentle stirring for the solution to react.

c. Titrate the solution with Sodium hydroxide to the two break points. The first being at a pH of approximately 5 and the second at a pH of 10. Record the volume at each point.

d. Repeat the titration using all the reagents except the sample in duplicate as blank determination.

Calculation:

$$\text{wt\% HAN} = \frac{(V_{1s} - V_b) \times N \times 9.603}{W_s} - 1.52 \times (\text{wt\% } HNO_3)$$

$$\text{wt \% TEAN} = \frac{(V_{2s} - V_{1s}) \times N \times 21.220}{W_s} - 2.65 \times (\text{wt\% AN})$$

Where:

W_S = weight of sample in gm.
V_b = Volume of titrant for blank (ml).
V_{1S} = Volume of titrant at first break point (ml).
V_{2S} = Volume of titrant at second break point (ml)
N = Normality of NaOH solution.
wt % AN = Weight percent ammonium nitrate as determined in 4.5.1.4.
wt % HNO_3 = Weight % Nitric acid as determined in 4.5.1.3

4.5.1.2 Water content. The water content shall be determined by the Karl Fischer method.

4.5.1.2.1 Equipment.

a. Aquameter - Beckman instruments, Inc. model KF2 or KF3 or equivalent. Equivalents are described in ASTM E 203.

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b. Microsyringe.

4.5.1.2.2 Reagents.

a. Hydranal Composite 5 (titrant) or equivalent.

b. Working Medium K or equivalent. Sources of Working medium K and Hydranal Composite 5K include Riedael-deHaen.

c. Hydranal water standard 1.0.

4.5.1.2.3 Procedure.

a. Standardize the titrant with the Hydranal water standard 10.0.

b. Fill a 100 µl microsyringe to 70 µl with the propellant solution and weigh to the nearest 0.1 mg.

c. Add 70 µl of the solution into the predried carrier and titrate to the end point using the Karl Fischer Aquameter.

d. Reweigh the microsyringe as soon as possible after the introduction of the sample.

Calculation:

$$\text{wt \% } H_2O = \frac{V \times T}{10 \times W_s}$$

Where:

V = Volume of titrant in ml.
T = Titer of titrant in mg water/ ml.
W_S = Weight of propellant solution in gm.

4.5.1.3 Nitric acid concentration.

4.5.1.3.1 Equipment.

a. Automatic titrator 716 DMS Titrino or equivalent.

b. pH meter.

4.5.1.3.2 Reagents.

a. Standardized Sodium hydroxide (NaOH) solution , 0.2 N (measured to four decimal places).

b. Standardized Nitric acid (HNO_3) solution, 0.25 to 0.3 N (measured to four decimal places).

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4.5.1.3.3 Procedure.

a. Dilute 28 gm of propellant with 50 ± 1.0 ml of distilled water.

b. Add 4 ml of the standardized 0.25 to 0.30 N Nitric acid solution to the sample.

c. Titrate to the end point with the standardized Sodium hydroxide solution.

Calculation:

$$\text{wt \% } HNO_3 = \frac{\left[(V_s \times N_{NaOH}) - (V_a \times N_a)\right] \times 6.301}{W_s}$$

Where:

V_S = Volume of titrant for sample (ml).
V_a = Volume of Nitric acid added (ml).
N_{NaOH} = Normality of Sodium hydroxide solution.
N_a = Normality of Nitric acid solution.
W_S = Weight of sample (gm).

4.5.1.4 Ammonium Nitrate concentration.

4.5.1.4.1 Equipment.

a. Micro-Kjeldahl distillation unit (figure 1).

b. Magnetic stirrer.

c. Micro-buret.

4.5.1.4.2 Reagents.

a. 1 % Starch solution - Triturate 5.0 gm of soluble starch with approximately 10 ml of deionized water. Pour into 500 ml of boiling deionized water. Allow to cool. Store solution in a refrigerator.

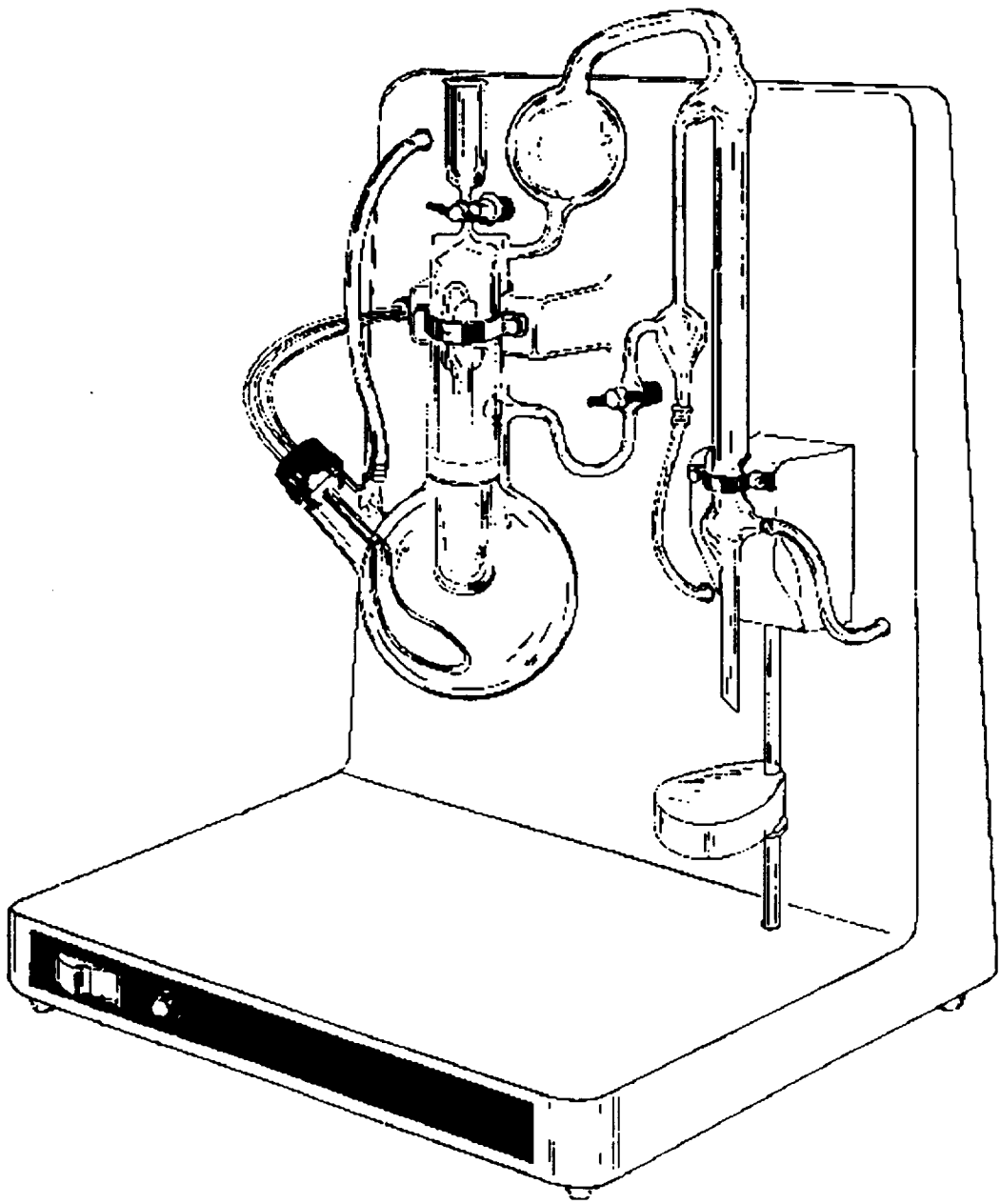
b. 20 % NaOH solution.

c. 2 % Boric acid solution.

d. 0.5 % methyl red indicator solution.

e. Iodine-Potassium Iodine mixture - grind 12.7 gm of Iodine with 40 gm of Potassium Iodine in mortar and pestle.

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MICRO KJELDAHL DISTILLATION UNIT

FIGURE 1

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f. Sodium bicarbonate.

g. Standardized 0.01 N Hydrochloric acid (HCl) solution (measured to four decimal places). Standardized using anhydrous Sodium Carbonate.

4.5.1.4.3 Procedure.

a. Deliver 1.0 ± 0.2 gm weighed to the nearest 0.01 mg of the liquid propellant into a 150 ml beaker, add 10 ml of deionized water and 3 gm of Sodium bicarbonate and 1 - 2 ml of the 1 % Starch solution.

b. With mixing, add small increments of the Iodine-potassium Iodine mixture until the solution remains black.

c. Transfer the contents of the beaker into the distillation flask quantitatively by washing with a minimum of deionized water. Deliver approximately 10 ml of the 20 % Sodium hydroxide solution to the flask.

d. Place a 150 ml beaker with approximately 20 ml of the 2 % Boric acid solution and five drops of the 0.5 % Methyl red indicator under the condenser outlet.

e. Perform the distillation and collect approximately 120 ml (total volume) of distillate in the beaker.

f. Titrate the 120 ml of distillate with the Standardized 0.01 N HCl solution to the end point. Measure the volume of HCl delivered.

g. Perform a blank determination with all reagents except the liquid propellant sample.

Calculation:

$$\text{wt \% AN} = \frac{(V_s - V_b) \times N \times 8.005}{W_s}$$

Where:

V_S = Volume of HCl for sample (ml).
V_b = Volume of HCl for blank (ml).
N = Normality of HCl
W_S = Weight of sample (gm).

4.5.1.5 Metals. The metals content of the liquid propellant shall be determined by Inductively Coupled Plasma mass spectroscopy (ICP-MS) (see 6.5). The value of total metals shall be the sum of the following metals in ppm: Chromium, Cobalt, Copper, Iron, Manganese, Mercury, Nickel, Vanadium and Zinc. In addition, the Mercury content shall be reported separately on a ppb basis and the Iron on a ppm basis.

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4.5.1.5.1 Preparation of sample for analysis. Dilute 10 ml of the propellant in 100 ml of a 2 % nitric acid solution.

4.5.1.6 Density. The density of the propellant shall be determined by pycnometer or density meter at 20 ± 0.1 degrees C.

4.5.1.7 Total nitrates. The total nitrates shall be determined by Ultra-violet absorbance.

4.5.1.7.1 Equipment.

a. Ultra violet spectrophotometer set to a wave length of 302 nanometers - Perkin Elmer model 330 Spectrophotometer or equivalent.

4.5.1.7.2 Reagents.

a. 0.1050 ± 0.0010 M Nitric acid (HNO_3) solution standardized using 0.2 N Sodium hydroxide. A 0.1050 ± 0.0010 M $NaNO_3$ salt solution may be used in place of the Nitric acid solution.

4.5.1.7.3 Procedure.

a. Determine the absorbance of the Nitric acid or the Sodium salt solution at 302 nanometers. Calculate the moles of Nitrate per absorbance unit.

b. Pipet 5.0 ml of the propellant into a preweighed 30 ml beaker and weigh to the closest 0.01 mg.

c. Transfer the contents of the beaker, washing thoroughly with deionized water, into a 500 ml volumetric flask. Dilute to volume with deionized water.

d. Determine the absorbance of the solution prepared above.

Calculation:

$$\text{wt \% NITRATE} = \frac{A \times C \times 310}{W_s}$$

Where:

A = Absorbance of the sample solution.
C = Calibration of instrument, molarity/absorbance
W_S = weight of sample in gm.

5. PACKAGING.

5.1 Packaging and packing.

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5.1.1 Level A. Not applicable.

5.1.2 Level B. Unless otherwise specified in the contract, packaging and packing shall be in accordance with the following drawings:

5 Gallon LP XM46 Pack - 12972309 and 12972308
30 Gallon LP XM46 Pack - 12972315 and 12972314

5.1.3 Level C. Shipments of laboratory samples of LP XM46 shall be as specified by the contracting officer.

5.2 Marking.

5.2.1 Level A. Not applicable.

5.2.2 Level B. In addition to any special marking required by the contract, each container shall be marked in accordance with the following drawings:

5 Gallon LP Pack - 12972309
30 Gallon LP Pack - 12972315

5.2.3 Level C. Shipments of laboratory samples of LP XM46 shall be as specified by the contracting officer.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The XM46 propellant produced in accordance with this specification is intended for use in the Advanced Fielded Artillery Systems (AFAS).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual document referenced (see 2.2).

c. Levels of packaging and packing (see 5.1 and 5.2).
d. Special marking, when specified.

e. Provisions for the submission of first article samples.

6.3 Submission of inspection equipment designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: SMCAR-QAR-R, Picatinny

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Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Equivalent test methods. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the contracting officer to Commander, ARDEC, ATTN: SMCAR-QAR-R, Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.5 ICP-MS. Sources of an Inductively Couple Plasma Mass Spectrometer include Perkin Elmer and Fisson.

6.6 Metals analysis. In addition to the metals called out in 3.3, the following metals shall be determined during the First Article test: Aluminum, Bismuth, Boron, Cadmium, Lead, Magnesium, Molybdenum, Palladium, Silver, Tin, Titanium, Tungsten, and Zirconium. If any metal on the list is found to have a concentration above 100 ppb, the contractor will be required to analyze for that metal during regular production. If it can be demonstrated by the contractor that by their process of manufacturing liquid propellant that there is not a path way for a particular metal called out in 3.1 and it is demonstrated that the concentration of the metal is below 100 ppb during the first article test, the analysis for that metal may be waived during regular production. A waiver will not be granted for Iron, Nickel or Copper.

6.7 Thermal stability. The thermal stability of the propellant shall be determined by accelerated rate calorimetry (ARC).

6.8 Sulfate ion concentration. If it can be demonstrated during first article testing that the sulfate ion concentration in the propellant is less than 2.0 ppm and that by the process of manufacture there is not a pathway for sulfate ions to be introduced into the propellant, the requirement for sulfate ions will be waived during regular production.

6.8.1 Equipment.

a. Ion chromatograph, Dionex DX-300 Series or equivalent with the following columns:

1. Anion trap column, ATC-1 Dionex or equivalent
2. Guard column, Ionpac AG10A-SC 4mm Dionex or equivalent.
3. Analytical column, Ionpac AS10A-SC 4mm, Dionex or equivalent.

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6.8.2 Reagents.

a. Nanopure water - The resistivity should be greater than 17.8 megaohm-cm.

b. Ultrapure Nitric acid (Sulfate concentration < 0.1 ppm).

c. 1000 ppm sulfate stock solution.

d. 1 % Nitric acid solution - Dilute 7.15 ml of the ultrapure nitric acid to 500 ml in the ultrapure water.

6.8.3 Preparation of eluent.

a. Stock eluent concentration (8 M NaOH solution) - dissolve 32 gm of ultrapure NaOH in 100 ml of filtered (0.45 µm), degassed water. To minimize the amount of carbonate formed in the NaOH, the pellets should be kept under helium and in a desiccator. The eluent should also be stored under helium.

b. Working eluent (80nM NaOH) - Pipette 10 ml of the stock eluent solution into a 1 L volumetric flask. Dilute to final volume with filtered, degassed water.

6.8.4 Preparation of standards. Prepare the following standards with the 1 % Nitric acid solution and the appropriate dilution of the stock solution.

Conc of Standard	ml of 1 % HNO_3	ml of Standard	Conc of Standard
0 ppm	4	0	-
0.25 ppm	2	2	0.5 ppm
0.5 ppm	2	2	1.0 ppm
1.0 ppm	2.4	1.6	2.5 ppm
2.5 ppm	2	2	5.0 ppm
5 ppm	4.98	0.02	1000 ppm
0.8 ppm QC	24.98	0.02	1000 ppm

6.8.5 Procedure.

a. Generate a concentration verses response curve based on the standards outlined in the table above. The operating conditions for the Ion chromatograph are as follows:

1. Eluent: 80 nM NaOH
2. Eluent flow rate: 1.0 ml/min.
3. Sample loop: 25 µl.
4. Run time: 35 min.
5. Suppresser: ARS-1 in recycle mode.
6. Detection: Suppressed Conductivity.

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b. Dilute a sufficient quantity of LP in water such that the concentration of NO_3^- in the sample is equivalent to that of a 1 % Nitric acid solution.

c. Determine the concentration of sulfate in the LP sample using the calibration curve generated above and applying the correction for dilution of the sample. Operating conditions for the instrument are the same as those used for the calibration procedure.

d. Calibration - Each time samples are analyzed with this method blank nitric acid must be run and if the background sulfate varies from the previous run, the calibration curve must be corrected for this. It is advised to have blank nitric acid samples mixed with test samples to monitor background sulfate. If it increases, the anion trap needs to be cleaned.

6.9 Submission of propellant description sheets and propellant acceptance sheets. In addition to the normal distribution of records, one copy of all propellant description sheets and propellant acceptance sheets shall be submitted to: ARDEC, ATTN: SMCAR-QAR-R, Picatinny Arsenal, NJ 08706-5000.

6.10 Ballistics. This specification presently contains no requirements for ballistic acceptance testing. This is due to the fact that a fixture to test the propellant does not exist. When one becomes available, requirements for ballistic acceptance testing will be incorporated into this specification.

6.11 Submission of alternate quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting officer.

6.12 Subject term (key word) listing.

Monopropellant
HAN
TEAN
Hydroxyl ammonium nitrate
Triethanol ammonium nitrate
Hydroxylamine nitrate

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1376-002)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-71158 (AR)	940826

3. DOCUMENT TITLE: PROPELLANT, LIQUID, XM46

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: AMSTA-AR-EDE-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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